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Building a Framework for the Digital Transformation of Italian Universities: Evidence From the University of Naples "Parthenope"

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ABSTRACT

This paper examines the digital transformation of universities, a category of knowledge-intensive organizations (KIOs) increasingly pressured to invest in new technological tools for the development of innovative teaching and research processes and practices. Despite the strategic relevance of digital technologies (DTs), prior studies highlight a persistent misalignment between their potential and their effective adoption within higher education. The objective of this study is to assess how universities can leverage DTs to enhance organizational effectiveness and to identify the main barriers that hinder their full integration. Adopting a single case study design, the research investigates the University of Naples "Parthenope." The empirical analysis combines documentary evidence with two questionnaires administered to academic staff ($N = 126$) and students ($N = 209$), aimed at capturing the current level of DT implementation and perceived opportunities and challenges. The empirical evidence shows that DTs are widely used to support routine teaching and administrative activities but are not yet exploited for more advanced, transformative purposes. Key opportunities relate to process digitalization, improved accessibility of services, and the development of more flexible and engaging learning models. Major barriers include organizational resistance, limited digital competencies among staff, technological fragmentation, and insufficient strategic alignment. By integrating theoretical insights and empirical findings, the study proposes a conceptual framework that systematizes the drivers, mechanisms, and obstacles shaping digital transformation in universities. While considering the typical limitations of the single case study, the study offers theoretical contributions to the literature on digital innovation in higher education and provides practical guidelines for university managers seeking to strengthen digital capabilities, promote cultural change, and support more advanced forms of technology-enabled innovation.

1 | Introduction

The evolutionary dynamics that redesign contemporary society are increasingly influenced by the changes dictated by the main Megatrends (Naisbitt 1982). Specifically, the term "Megatrend" describes "*large social, economic, political, and technological changes (...), they influence us for some time—between seven and ten years, or longer*" (Naisbitt 1982, 9). This concept, therefore, identifies disruptive forces that can affect consumer behavior, government policies, and technological evolutions by redesigning the business landscape (Lee et al. 2019; Gajdzik et al. 2021; Jeflea et al. 2022). Currently, technological disruptions driven by self-learning robots, extended reality, and the Metaverse, as well as cyber warfare and security, represent some of the key issues expected to influence the global economy and business practices in the near future (EY 2025; IFF 2025; PwC 2025). Overall, these developments appear to converge into the broader process of digitalization. Both theoretical and empirical studies consistently frame digital transformation—and the increasing adoption of digital technologies (DTs)—as a set of

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Summary

This paper analyzes how digital transformation unfolds within universities, focusing on the opportunities offered by digital technologies (DTs) and the barriers that hinder their full integration. Specifically, it:

- Investigates the role of digital transformation in universities, focusing on how digital technologies reshape teaching, organizational processes, and strategic orientation.
- Examines the digital transformation journey of the University of Naples "Parthenope" through a qualitative single-case study, combining documentary analysis with survey data collected from professors and students.
- Proposes a conceptual framework that integrates technological, organizational, cultural, and ethical dimensions to explain digital transformation in higher education institutions.
- Highlights that DTs are widely used for routine activities, while their transformative and advanced applications remain limited.
- Provides theoretical and managerial implications by outlining strategies to foster alignment between infrastructure, skills, culture, and governance in universities.

socio-technical processes that affect individuals and organizations, capable of triggering substantial changes in business models, competitive strategies, and organizational structures, as well as in the underlying logics and assumptions that guide corporate value-creation processes (Vial 2019; Capurro et al. 2024).

In recent years, indeed, studies about digital transformation and the related DTs have stimulated particular interest in the scientific and professional community (Plekhanov et al. 2023; McKinsey 2023; Deloitte 2024). The rapid technological progress has offered the possibility to achieve business' processes improvements as well as strategic and organizational models' transformations in terms of innovation and sustainability (Caputo et al. 2021). The diffusion and adoption of DTs have had a profound and pervasive impact on society, influencing sectors that until recently were still resistant to digital transformation, including knowledge-intensive organizations (KIOs) and universities (De Nito et al. 2023; Van Meeteren et al. 2022; Mercader and Gairín 2020).

Nowadays, universities from all over the world are facing the challenges posed by digital transformation in order to adapt to the rapid external changes and to maintain high competitiveness at both the national and international levels. The implementation of DTs drives the change process, redesigning conventional university models and practices for teaching, research, and third mission activities as well as for administrative processes (Richards et al. 2004; Starkey 2020; Caputo, Hack-Polay, et al. 2024).

Several universities worldwide have launched initiatives aimed at modernizing their digital infrastructures and strengthening the digital skills of both academic and administrative staff (Zarubina et al. 2024; Antonopoulou et al. 2023; Quy et al. 2023). These initiatives often include investments in new technological tools, the development of innovative teaching methodologies, and targeted training programs to support digital competence building (Mhlanga 2023; Ali et al. 2024). At the policy level, governments and international institutions have also promoted strategic plans and allocated dedicated financial resources to accelerate the digital transition of higher education systems (i.e., National Recovery and Resilience Plan—PNRR). Such interventions are intended to enhance universities' capacity to fully exploit the opportunities offered by DTs and to address the emerging challenges associated with digital transformation.

Despite their rapid diffusion and relevance, previous research on DTs (Alnagrat et al. 2022; Wang et al. 2024; Mexhuani 2025) shows resistance and slowness in their implementation and exploitation by universities, and to date, they are principally used in the simplest and most basic forms. This delay has prompted scholars to investigate how universities are exploiting the potential offered by DTs and the difficulties that hinder their full integration (Deacon et al. 2025; Scholkmann et al. 2024).

In light of the identified gap, the research aim is to examine how universities can effectively leverage the opportunities presented by DTs to promote more innovative, sustainable, and inclusive strategies, models, and organizational practices. Additionally, the research seeks to analyze the resistance and barriers that hinder their implementation and identify strategies to overcome these challenges.

To support the research objectives, the study first develops a literature review about the topic of digital transformation and its impact on universities. The analysis of the existing literature is enriched through the single case study methodology (Yin 2009). In particular, the research intends to investigate the current use of DTs within the University of Naples Parthenope, to explore how the Institution could further exploit the potentialities offered by DTs to improve attractiveness, competitiveness, and both didactic and management efficiency, and, finally, which are the current resistances that hamper their full integration. Moreover, case study analysis is supplemented by the administration of two questionnaires addressed to the University's professors and students to collect relevant information regarding the current level of implementation of DTs.

From an analysis of literature and from the empirical data collected through the case study and the questionnaires, a conceptual framework is developed to highlight, on one hand, the opportunities and solutions related to the implementation of the new technologies and, on the other hand, the barriers to their integral assimilation within the higher education sector.

This study advances research on strategic management and digital innovation by offering an examination of the drivers, mechanisms, and constraints that shape digital transformation within higher education institutions. Through the analysis of a single case study, the study clarifies how DTs can reconfigure organizational processes and strategic orientations while also revealing the structural and cultural barriers that limit their full integration. By systematizing these elements, the study provides

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a conceptual framework that can support future theorization and guide comparative investigations on digital transformation across universities.

The paper is organized as follows: Section 2 is dedicated to the literature review; Section 3 describes the methodology implemented; Section 4 highlights the results emerged from the case study; Section 5 is dedicated to the discussion of the findings; Section 6 provides the conceptual framework; and finally, Section 7 offers the main conclusions, contribution to literature, managerial implications, and suggestions for future research.

2 | Literature Review

Business-economics doctrine is currently committed in studying the constantly evolving socio-economic environment in which companies compete, in order to understand the determinants concerning the dynamics and deep transformations that they must face (Chesbrough 2010; Yang et al. 2023; Patrucco et al. 2020; Fiorentino et al. 2020). The continuous stimuli derived from technological innovations require companies to adopt a dynamic and proactive strategic approach in order to grasp and anticipate needs and trends as well as identify new sources of competitive advantage (Porter 1985; Garzella et al. 2020; Capurro et al. 2024; Leão and da Silva 2021).

The advent of digital transformation has, in fact, radically changed society and the way in which companies create and distribute value, causing a radical change in sector boundaries and dynamics (Verhoef et al. 2021; Garzella et al. 2021).

Digital transformation, together with the increasing adoption of DTs, has completely redefined the competitive, financial, and social dynamics of companies. Specifically, the term DTs includes a wide range of automated systems—including Internet of Things, Big Data Analysis, Blockchain, Artificial Intelligence (AI) together with the assets identified by the industry 4.0—characterized by "Self-Monitoring, Analysis and Reporting Technology (SMART)" (Mashhadi et al. 2018, 1108). Indeed, the introduction of these new technologies has triggered a radical process of digital transformation that has invested almost all companies (Fiorentino et al. 2020; Caputo et al. 2022). This change has prompted considerable interest among scholars and practitioners, leading to a substantial increase in investment to understand and test the impact of DTs, even in knowledge-intensive sectors characterized by resistance to transformation, which include universities (De Nito et al. 2023; Van Meeteren et al. 2022; Mercader and Gairín 2020).

Thus, DTs emerged as a transformative force in several sectors, and the educational one is no exception (Beardsley et al. 2021; Patrício and Ferreira 2024). The education institutions—especially those of high grade—have experienced profound change in recent years as rapid advances in DTs have reshaped both teaching and organizational processes (Williamson and Eynon 2020; Altassan 2025). Digital tools such as learning management systems, online platforms, and mobile applications have transformed students' learning experiences and faculty–student interaction, moving education toward more flexible, engaging, and student-centered approaches (Dhawan 2020; Mishra et al. 2020; Kalyani 2024). Universities face unique challenges and opportunities in this process. Higher education institutions must balance the adoption of advanced technologies with the core mission of teaching, research, and societal engagement (Selwyn 2014; Henderson et al. 2015; Wang et al. 2024).

The COVID-19 pandemic further accelerated this transition, revealing both the strategic importance of robust digital infrastructures and the need for coherent digital strategies to ensure continuity and quality in education (Skulmowski and Rey 2020; Amankwah-Amoah et al. 2021). The emergency and the provision imposed by the State to guarantee the ongoing progression of academic activities forced universities to undertake a rapid path of digital development (Bond et al. 2018; Ceravolo et al. 2023; Bianchi et al. 2025).

Overall, DTs have become essential to strengthening institutional effectiveness, supporting strategic decision-making, and fostering innovation within education institutions (Wekerle et al. 2022; Mexhuani 2025).

DTs' introduction in the university system, to date, not only represents the willingness to maintain inter-university competitiveness high, but also the need to comply with national and international standards and directives in order to improve the quality of teaching and learning processes, research and social development as well as to ensure a correct and ethical use (UNESCO 2023; OECD 2023; European Commission 2020; Governo Italiano 2021).

For universities, digital transformation is no longer a choice, but rather an inevitable progress aimed at preserving its relevance in the higher education sector (Nguyen 2018; Bianchi et al. 2025). The implementation of DTs enables the improvement of the overall teaching and learning experience (Ratten 2023; Henderson et al. 2017), offers new tools for research, facilitates the establishment of strategic relationships between universities and external actors promoting socio-economic development (Dell'Atti et al. 2023; Rippa and Secundo 2019), and allows for simplify administrative practices and the increase of sharing and transparency of information (Ahmad et al. 2022).

The opportunities offered by DTs are considerable, but the process of their integration and implementation has met—and still meets—several resistances (Gkrimpizi et al. 2023; Aditya et al. 2022). Different studies (Müller and Leyer 2023; Bond et al. 2018; Henderson et al. 2017) indeed highlighted how the implementation of DTs is still limited to the use of simple and basic forms such as multimedia tools, institutional e-mail, or e-learning systems. In literature, this underutilization is recognized by several factors, including the lack of precise institutional plans for integrating DTs into university activities, the low level of digital knowledge among university staff (academic and administrative), and the obsolescence of structures for integrating new technologies into institutional activities (Mercader 2020).

Currently, literature recognizes digital transformation in the university context as a relevant socio-economic challenge that

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requires coordinated development between people and technologies to be successfully addressed (Ahmad et al. 2022).

The transformative process dictated by the adoption of DTs is not limited to the introduction of such smart technologies and/or tools in the various institutional activities but also includes the need to develop an organizational culture oriented toward change and innovation, emphasizing the importance for universities to adopt an entrepreneurial orientation (Caputo, Pacheco, et al. 2024).

Human capital plays a key role, both at governance level—which drives adoption and implementation—and operational level—where human capital concretely uses technology to improve and increase operational efficiency and effectiveness (Verhoef et al. 2021).

In the higher education sector, these objectives can be achieved by placing people at the center of all university processes, involving the entire reference community, such as students, professors, and technical-administrative staff.

Therefore, despite the widespread consensus on the strategic relevance of digital transformation in higher education, the literature reveals several underlying tensions and contradictions. On the one hand, scholars emphasize the potential of DTs to enhance learning quality, organizational efficiency, and institutional competitiveness; on the other, empirical evidence consistently shows fragmented adoption, misalignment between technological investments and pedagogical practices, and persistent cultural resistance within academic communities. This contrast highlights a structural paradox: while universities are urged to embrace digital innovation as a catalyst for modernization, their organizational logics, governance structures, and professional identities often remain anchored in "traditional" models, limiting the transformative scope of digital initiatives (Zarubina et al. 2024; Antonopoulou et al. 2023).

These aspects suggest that the actual transformative capacity of DTs within universities remains far from linear or guaranteed. The coexistence of strong institutional pressures toward digitalization with persistent organizational, cultural, and pedagogical constraints indicates the need for deeper investigation into how digital transformation unfolds in practice.

In light of these, the study poses the following research questions: (1) *What is the impact of digital transformation and the related DTs within universities?* (2) *Which opportunities can universities strategically leverage to develop stimulating, innovative, and participatory teaching and learning processes?* (3) *Which challenges and barriers limit the full implementation of DTs?*

3 | Research Methodology

3.1 | Case Study

To achieve the research aim, the paper adopts the single-case study methodology (Yin 2009, 2016; Yin and Campbell 2018; Gustafsson 2017), identifying the University of Naples Parthenope (hereafter referred to as Parthenope or the University) as the empirical business case.

According to Yin (2009), the adoption of a single case study design is appropriate when the case can be considered critical, revelatory, unique, or emblematic with respect to the phenomenon under investigation. In this study, the choice of Parthenope University is justified on multiple, theoretically grounded criteria.

First, the case can be classified as a critical case, as it enables the testing of theoretical assumptions in contexts where the phenomenon is expected to occur but may be limited or uneven. Parthenope University provides an ideal setting to investigate whether and how digital technologies can facilitate meaningful transformation in teaching and organizational practices, despite the strong institutional, regulatory, and resource constraints that characterise many universities.

Second, the case can also be interpreted as emblematic (Yin 2009; Gustafsson 2017), as it reflects broader dynamics affecting contemporary universities rather than representing an extreme or exceptional situation. With a century-old tradition in education, Parthenope University is considered a high-level education center focused on the advancement of innovative, international, and multidisciplinary research, open to comparison with real-world application experience and the world of work and business. It promotes the growth of both didactic and scientific skills of professors and researchers and the cultural and professional training of students; it also fosters the direct application, valorization, and use of knowledge to contribute to the social, cultural, and economic development of society (Parthenope University website). Thus, Parthenope University is a medium-sized public university with a long-standing academic tradition and a diversified disciplinary structure, making it representative of the organizational complexity, governance challenges, and strategic tensions faced by many higher education institutions undergoing digital transformation. As such, the case provides an illustrative example of how digitalization pressures are translated into concrete strategies, initiatives, and practices within universities.

Third, the case has a revelatory value, as it allows in-depth access to internal strategic documents, governance processes, and stakeholders' perceptions at a critical moment in the institution's transformation journey. Yin (2009) highlights that revelatory cases are particularly valuable when they offer the opportunity to observe and analyze phenomena that are otherwise difficult to access. In this study, the availability of strategic plans, internal reports, and direct feedback from professors and students enables a detailed exploration of the mechanisms, opportunities, and resistances shaping digital transformation in higher education.

The relevance of the case is further reinforced by the University's explicit strategic commitment to digital innovation. Recently, indeed, the University has embarked on a significant transformational process in response to the innovative changes affecting our society today. The University Strategic Plan 2022–2028 emphasizes the need to address emerging societal and technological challenges, recognizing the university's role in shaping future educational models: "The process of developing the university strategy requires the identification of concrete emerging

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challenges, in the awareness that the university has a role to play in shaping and building the future." (Strategic Plan 2022–2028, 5). In line with this strategic orientation, the Plan identifies specific action lines focused on innovative teaching methodologies and the professional development of academic staff.

To support these objectives, a dedicated working group was established in October 2023 to explore innovative teaching practices and to map professors' needs and preferences regarding digital tools and methodologies.

Therefore, the selection of Parthenope University is not based on its historical prestige per se, but on its suitability as a theoretically informed case that allows for the exploration of digital transformation as a socio-technical and organizational process within higher education. Consistent with the selected methodological approach, the purpose of the single case study is not statistical generalization but analytical generalization, whereby the findings contribute to theory refinement and the development of a conceptual framework that may inform future comparative and multi-case research on digital transformation in universities.

3.2 | Data Collection

In order to study the case, a specific protocol of analysis was followed (Yin 2016). After constructing a basic theory, the authors moved to data collection, which was conducted combining primary and secondary sources of data, which allowed us to triangulate different sources effectively. In order to increase the reliability of our results, data collection were carried out using the following sources: document analysis and qualitative questionnaire.

The sources of evidence were selected and used like a chain (Yin 2016), with an explicit link between the theoretical background, the examination of documents, the choice of questionnaire subjects, and the formulation of questionnaire questions.

Specifically, data collection was carried out using a multimethod approach (Myers 2013). Firstly, a documentary analysis was conducted using reports available on the University website (https://www.uniparthenope.it/).

In particular, the main information was obtained by studying the following sources:

- University Strategic Plan 2016 2017, 2018–2021, 2022, 2023, 2024a and 2024b, 2025-2028.
- Working Group Report N. 1/2024, Recognition Activities on Innovative Didactics and Teaching Learning Initiatives.
- Three-year Information Technology Plan 2022–2024.
- Integrated Activities and Organization Plan 2024–2026.
- General University Regulations.

Subsequently, based on the information gathered and the research gap identified, a first questionnaire was elaborated to be addressed to the University's professors, with the aim of emphasizing the typical elements and aspects of teaching activities and the current degree of adoption of innovative teaching methods and digital technologies. A second questionnaire, in line with the topic of the first, was addressed to the students with the aim of validating the data collected from the first one and testing the students' inclination and willingness to use new and innovative teaching methodologies.

Both questionnaires were administered online via Google Forms, structured in an anonymous form, over a period from October 2023 to April 2024, with a maximum time of 15 min to complete the questions.

Specifically, the professors questionnaire consists of 15 questions, 5 of which require the filling of a space with text and the other 10 are multiple choice or binary yes/no questions; while the students questionnaire consists of 12 questions, 4 of which require the filling of a space with text, 6 are multiple choice or binary yes/no, and 2 are presented in the form of Likert scales ranging from 1, "Absolutely no," to 5, "Absolutely yes." For greater clarity and transparency, the full versions of the questionnaires administered to professors and students are provided as Supporting Information.

3.3 | Data Analysis

The data collected through document analysis and questionnaires were analyzed using a qualitative approach. Specifically, documentary sources were analyzed through content analysis (Eisenhardt 1989) to identify relevant themes and institutional priorities related to the digital transformation process.

The questionnaires were then analyzed in order to integrate and validate the documentary findings, thus allowing for the identification of recurring patterns related to the adoption, use, and perception of DTs within the University.

In accordance with the principles of triangulation (Yin 2016; Jonsen and Jehn 2009), the integration of multiple data sources facilitated a more comprehensive and reliable understanding of the phenomenon under examination. The analysis was guided by both deductive and inductive approaches (Graebner et al. 2012): the former, through the use of theoretical constructs drawn from the extant literature on digital transformation in higher education; the latter, instead, through the emergence of themes derived directly from empirical evidence.

The categories and themes that emerged from this process had a significant impact on the development of the conceptual framework proposed in the final section of the paper.

The main findings resulting from the analysis of the questionnaires are presented in the next section.

4 | Results

4.1 | Results From the Professors' Questionnaire

The survey was addressed to all professors to map the adoption of innovative teaching methods and the use of DTs, as well as to identify any preferences and/or difficulties encountered while teaching activities.

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The response rate of 36% (126 respondents out of a total of 346 structured professors) is appropriate to represent the perception of Parthenope University's professors on the issues, in line with previous studies on the subject (Ogier 2005; Nulty 2008). The distribution by academic role is rather homogeneous (27% full professor, 33% associate professor, and 40% assistant professor).

The first question was carried out to identify the problems and critical issues that professors face when running courses (Figure 1). Respondents reported several recurring problems related to the classroom experience that may be related to the issue of student engagement, such as discontinuous attendance and/or high dropout rates; low attention/concentration and difficulty in involving students in classroom discussions; low student participation in teamwork; lack of incentives for students to attend and participate actively; and inadequate facilities and classrooms.

In addition, respondents were asked whether they employed innovative teaching methods during lessons, understood as the possibility of using teaching methodologies that are alternative and/or supportive to frontal lectures and if they used DTs to implement innovative teaching methods.

In both cases, the most responses were largely positive (74% for innovative teaching methods and 84% for the use of DTs to support innovative teaching).

However, on closer inspection, the main innovative teaching methods used by respondents continue to be traditional case studies (75.2%) and group activities (61.4%) (Figure 2).

There is also a tendency toward traditional tools (Figure 3) such as the use of multimedia resources (videos, images, etc.) to make learning content more engaging; tools to support remote learning such as Zoom, Microsoft Teams, and Google Meet; E-learning (online courses, tutorials, and digital learning resources); Learning Management Systems as online platforms that allow the organization, distribution, and management of learning materials, assignments, and communication with students (Moodle, Blackboard, Canvas, and Google Classroom).

The use of more advanced DTs that enable personalized, immersive and interactive learning experiences, such as virtual reality (VR) (0.8%) and augmented reality (AR) (0.8%), AI (1.6%), learning analytics, that is, the use of data to monitor and improve student progress (3.2%), robots for teaching

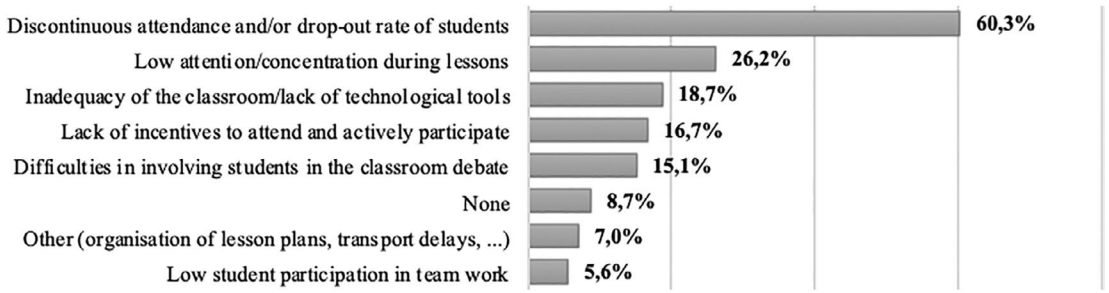


FIGURE 1 | Main problems identified in classroom activities. *Source:* Own elaboration.

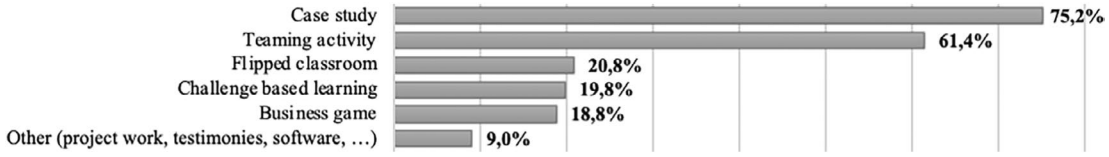


FIGURE 2 | Types of innovative teaching methods employed. *Source:* Own elaboration.

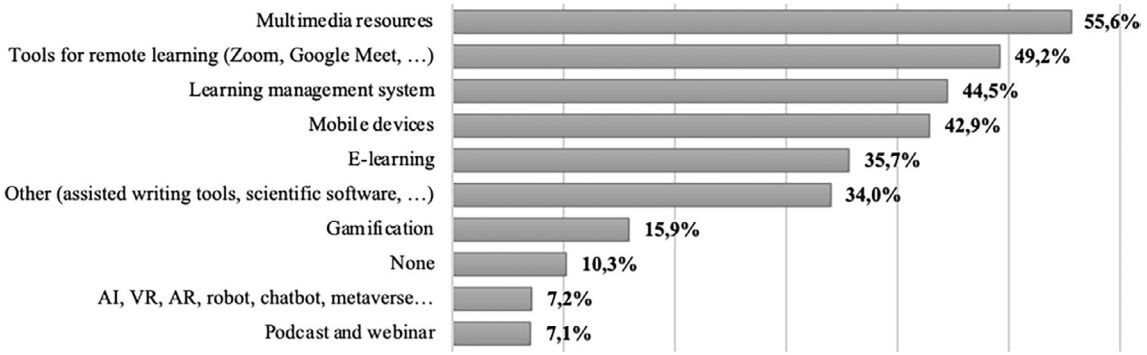


FIGURE 3 | Types of digital technologies implemented to support teaching activity. *Source:* Own elaboration.

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(0.8%), chatbots and virtual assistants (0%), metaverse (0%), is low or absent.

In general, therefore, the results show a high propensity to use innovative teaching methodologies and DTs to support them, but only in the simplest and most basic forms.

There is considerable potential to expand the use of active and student-centered learning approaches supported by innovative digital tools, especially as most respondents said they would use new innovative teaching methods and DTs in the next year (79%).

4.2 | Results From the Students' Questionnaire

The survey was extended to the University's students. The purpose is to validate the answers obtained from the questionnaire administered to the professors and to find out their opinion on the use of innovative teaching methodologies and DTs, as well as to identify any preferences and/or difficulties encountered in the classroom experience.

The data collected allows a preliminary assessment of students' propensity and willingness to use new and innovative teaching methods.

Regarding the classroom experience, as observed in the professors' questionnaire, the students' main criticisms were the professors' difficulty in involving students in class discussion, the lack of incentives to attend and actively participate in lessons, the inadequate classroom structure and tools, and the lack of practical activities.

Respondents were then asked whether they had been provided with innovative teaching activities and which tools had been used. Some respondents (41.3%) affirm that they had not used innovative teaching tools. Those who said they had (58.7%) confirmed the methods that had emerged from the professors' questionnaire.

Respondents were then asked whether they had used DTs. Despite the positive response rate, traditional tools remain the most widely adopted.

The student questionnaire, therefore, fully confirms the professors' responses. This correspondence provides important information about the limitations of current teaching activities and the use of DTs; it also shows which areas should be developed to improve teaching efficiency in a university context.

5 | Discussion

The analysis of the questionnaire responses highlights significant elements regarding the adoption of innovative teaching methodologies and the use of DTs. While the literature review highlighted potential opportunities, the case study provides context-specific insights into how these dynamics manifest within the examined institution. In line with the methodological characteristics of single-case studies (Yin 2009), these findings do not allow for generalization to the Italian university sector as a whole; rather, they offer an analytical perspective that helps uncover patterns of opportunities, limitations, and resistances that may also emerge in similar institutional contexts.

Specifically, in line with the aim of this study, the discussion addresses the three research questions, integrating the evidence emerging from the literature review with the insights obtained through the case study and the analysis of the questionnaire responses.

The first research question concerns the impact of digital transformation and related DTs on university activities. The analysis shows that digital transformation has generated significant, although uneven, effects on the academic context. The case study confirms that the COVID-19 pandemic represented a critical turning point; at Parthenope, it accelerated the adoption of a centralized e-learning strategy, increasing the supply of online courses and enabling remote learning modalities. However, the integration of DTs within face-to-face teaching remains more limited, producing only a modest impact on classroom engagement.

Despite this partial implementation, the questionnaires reveal that professors have developed a strong awareness of the need to innovate their teaching methods, recognizing digital tools as strategic levers to enhance teaching effectiveness. These findings align with the literature, which highlights the transformative potential of DTs in fostering more interactive, personalized, and engaging learning processes (Hwang and Chien 2022; Viberg et al. 2024).

"Digital technologies have significantly changed the way I interact with students, allowing for more personalized learning paths and greater engagement, both during and beyond classroom activities." (Professor, survey respondent).

"The adoption of digital technologies has made teaching more flexible and responsive. This has increased participation and improved the overall learning experience."

(Professor, survey respondent).

For instance, AI is already widely used in higher education, with evidence showing improvements in classroom management, personalized learning, and continuous optimization of teaching processes (Wang et al. 2023; Viberg et al. 2024). Likewise, the Metaverse offers immersive and interactive environments that can enhance experiential learning, gamification, and global collaboration (Lin et al. 2022; Yao et al. 2024).

Overall, digital transformation has introduced both organizational and pedagogical changes, opening the way to a rethinking of traditional teaching approaches. However, its actual impact appears highly dependent on institutional strategies and the willingness of professors to integrate innovative tools into everyday teaching practice.

"Digital technologies are a valuable support for teaching; however, their everyday integration remains uneven across

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disciplines, with more traditional fields, such as law and accounting, being less easily adaptable to digital tools."

(Professor, survey respondent).

The second research question focuses on the opportunities that universities can strategically leverage to develop stimulating, innovative, and participatory teaching and learning processes. The findings highlight a wide range of opportunities, many of which emerge clearly from the alignment between the literature and the empirical evidence (Wekerle et al. 2022; Mexhuani 2025). Digital technologies provide new possibilities to make the classroom experience more dynamic and less centered on traditional information transmission. Innovative methodologies, supported by digital tools, foster interactivity and active learning, thereby reducing student passivity.

"One of the most positive aspects I have experienced is the opportunity to interact with other students beyond the traditional classroom setting. The relationship with professors has also improved, allowing for the exchange of experiences and reflections, especially when working in small groups."

(Student, survey respondent).

"It is certainly an engaging and effective methodology for learning theoretical concepts within a more practical and dynamic context, while also helping to maintain attention and enhance enjoyment of the lecture."

(Student, survey respondent).

The case study also shows that professors not only acknowledge these opportunities but also express a positive attitude toward adopting such methods. Overall, digital transformation enables the development of a more personalized, inclusive, and engaging educational ecosystem, where the student experience becomes central to the learning process.

"I view digital transformation as an opportunity to redesign teaching in a more student-centered way, making learning paths more flexible and inclusive."

(Professor, survey respondent).

"Technology helps reduce barriers to participation, making teaching more accessible to students with different backgrounds and learning styles." (Professor, survey respondent).

However, while the questionnaires reveal important opportunities to rethink and deeply innovate the educational experience through DTs, they also highlight significant limitations and obstacles to fully exploit their potential.

Indeed, the third research question addresses the challenges and barriers that hinder the full implementation of digital technologies within universities. Both the literature and the empirical evidence indicate the presence of significant obstacles that span technological, economic, organizational, and ethical dimensions (Gkrimpizi et al. 2023; Aditya et al. 2022). One of the main concerns relates to data management and privacy, as AI and Metaverse platforms often collect large volumes of sensitive information about students' behaviors and learning habits.

"AI and immersive platforms generate large amounts of sensitive data, and it is not always clear who has access to this information or how it is ultimately used."

(Professor, survey respondent).

"While digital platforms improve learning, I am concerned about how much personal information is being collected about my study habits and online behavior."

(Student, survey respondent).

The adoption of such technologies also poses significant challenges in terms of implementation costs, as they require significant investment in advanced technological infrastructure—such as servers, software, interactive devices, and virtual reality environments—as well as the necessary presence of developers, specialists, and significant costs on in-house training of professors and administrative staff (Cirillo et al. 2023; Uzule and Verina 2023). For universities, which often operate with limited budgets and resources for multiple purposes, these investments can be relevant barriers.

The effective implementation of digital technologies depends heavily on in-house training, but allocating time and financial resources to upskill staff remains a major challenge.

(Professor, survey respondent).

Beyond technological investments, there is a strong need for specialized personnel and continuous training for both academic and administrative staff, which represents a significant cost for the institution.

(Professor, survey respondent).

Digital inequality represents an additional issue; not all students have access to suitable devices or stable internet connections, which may prevent certain groups from fully participating in immersive or personalized educational experiences.

"When designing digital or immersive activities, we must consider that not all students have equal access to technology, which can unintentionally increase educational inequalities."

(Professor, survey respondent).

"Online and digital learning assume that everyone has the same technological resources, but this is not always the case, especially for students from disadvantaged backgrounds."

(Student, survey respondent).

Moreover, there is a broader fear of a possible "dehumanization" of education, as increasing reliance on technology may reduce interpersonal interaction and weaken the emotional and social

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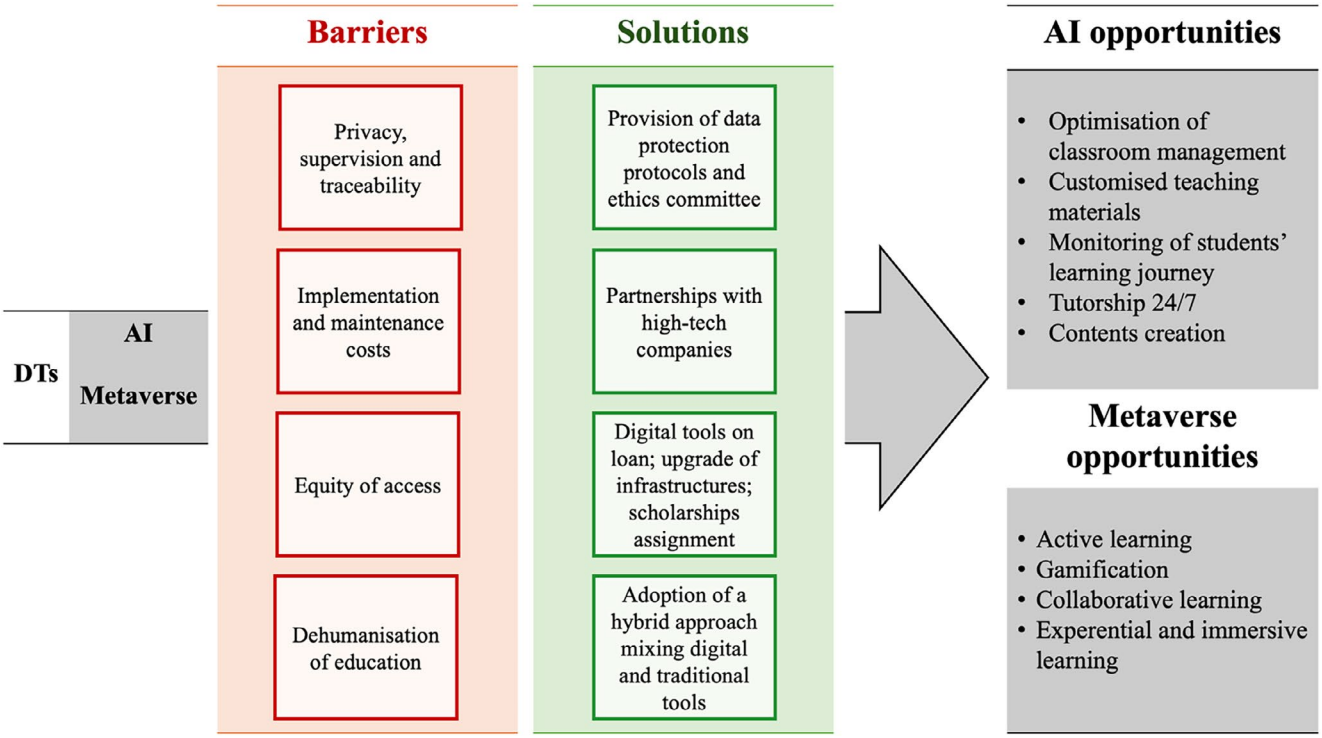


FIGURE 4 | A conceptual framework for the higher education sector. *Source:* Own elaboration. [Color figure can be viewed at wileyonlinelibrary.com]

dimensions that have traditionally characterized the learning process.

"My concern is that an excessive reliance on digital tools may reduce face-to-face interaction, weakening the relational and emotional dimensions that are essential to the educational process."

(Professor, survey respondent).

Overall, from the case study analysis emerges that professors still face obstacles related to limited institutional support, insufficient training, and the difficulty of achieving a balanced integration between traditional teaching methods and digital tools.

Taken together, these barriers should not be interpreted solely as limitations but rather as essential starting points for understanding how to guide future innovation strategies. While digital technologies offer the potential to profoundly rethink teaching and learning processes, their adoption must be approached in a balanced, ethical, and sustainable manner. To promote a more engaging and effective learning experience, it is crucial to introduce digital tools that allow for a balance of traditional and innovative teaching methods (MIM 2022). Only through a conscious and well-supported implementation of digital transformation will universities be able to build an innovative and participatory learning environment that meets the evolving needs of students.

6 | A Conceptual Framework for the Higher Education Sector

Understood from the literature the potential of DTs and the multiple barriers to their full integration in the higher education sector—and having verified the current level of technology and the propensity for improvement through the empirical analysis of the case study—the research aims to propose a conceptual framework that can be useful for promoting the implementation of the DTs (Figure 4).

The conceptual framework aims to support practitioners and academics in taking advantage of the opportunities offered by DTs and to provide potential solutions to the barriers that arise through ideas and suggestions that can potentially be replicated by all universities. The framework is not intended to be exhaustive or complete, but rather it encourages researchers and practitioners to improve the solutions offered, to propose new ones, and to integrate empirical evidence.

The framework will therefore focus on the barriers outlined above, recognizing that there may be others.

In addressing the challenges related to privacy and the traceability of students and professors, a potential solution involves the development of educational platforms that incorporate precise data protection protocols (Alier et al. 2021). These platforms should prioritize the minimization of data collection, limiting it exclusively to what is necessary for platform enhancement and the resolution of potential bugs. Additionally, these platforms must comply with all relevant data protection regulations (Komljenovic 2022). The establishment of a special independent body, an ethics committee for compliance with data collection and use, to oversee the proper functioning and operation of the educational platform would be of significant benefit. The establishment of a committee composed of experts and professionals from outside the university and independent of it would ensure greater trust and transparency in the use of the educational platform (Isus et al. 2024).

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The high costs of implementation, operation, and maintenance are certainly an important barrier, and the dimension may vary according to the size of universities (Benke and Widger 2023). It seems clear that, at least in the current scenario, the State resources allocated to universities are limited and there is a tendency to use them to satisfy other types of needs. Thus, universities should follow different paths to be able to develop and implement DTs. Among the various possibilities, one solution could be found in the external development strategy (Østergaard and Drejer 2022; Garzella et al. 2021) by entering partnerships with high-tech companies in order to have proprietary platforms created *ad hoc* at favorable prices.

More economical alternatives could include: the choice of platforms and tools already in use in higher education, which are certainly cheaper than proprietary platforms, but the choice of standardized products may not fully reflect the needs of a particular university; the choice of free or low-cost open-source platforms and tools, but with limited and standardized functionality (Alenezi 2021).

In terms of ensuring equity of access to technology for students, this is not a "new" issue in the Italian university system. Several initiatives have already been taken to bridge the technological and social gap between students. These initiatives concern: the distribution of technological devices on loan for use by students (Ali et al. 2024); the upgrading of university infrastructures through the creation of new study areas equipped with hardware and software devices that are freely accessible to students; and the assignment of scholarships awarded based on the academic results achieved and the socio-economic status of the students.

Finally, a massive and unregulated use of DTs could compromise human relationships and limit the social benefits of active participation in university life (Thornberg et al. 2022). To mitigate these risks and prevent technology from limiting social interaction, it is important to adopt a hybrid approach that harnesses the power of DTs without completely replacing traditional teaching methodologies.

To achieve this objective, academic institutions must establish precise guidelines regarding the utilization of the available tools, as well as the circumstances and methods by which they should be employed (Costa et al. 2022). Furthermore, professors must receive training in innovative teaching methodologies, as well as the integration of digital and traditional tools to optimize the benefits of both.

In conclusion, the paper aims to offer a conceptual framework that is designed for the implementation of DTs in universities. Starting from the opportunities that can be exploited to improve university performance from different perspectives (teaching quality, student satisfaction, professor satisfaction, etc.), the framework aims to provide solutions to overcome the barriers that arise when trying to leverage the potential of DTs.

7 | Conclusion

This study set out to investigate how digital transformation unfolds within universities, focusing on the opportunities offered by DTs and the barriers that hinder their full integration. Adopting the single-case study method, the research examined the digital transformation journey of the University of Naples "Parthenope" through a qualitative approach, combining documentary analysis with a survey addressed to academic staff and students.

The findings provide a nuanced and empirically grounded understanding of the current state of digital transformation in higher education, highlighting both its enabling potential and its structural, organizational, and cultural constraints. The survey conducted offers an analytical perspective that helps uncover patterns of opportunities, limitations, and resistances that may also emerge in similar institutional contexts.

Overall, the results show that digital technologies are widely adopted to support routine teaching and administrative activities, while their use for more advanced, transformative, and immersive purposes remains limited. Although professors and students express a generally positive attitude toward digital innovation, the integration of DTs into everyday teaching practices is still uneven and largely dependent on disciplinary characteristics, individual initiative, and institutional strategies. However, the findings emerging from the case study reveal encouraging trends, indicating the willingness of both students and professors to integrate these new tools more extensively and promptly to facilitate and accelerate knowledge exchange and enhance the overall quality of teaching. These aspects confirm that digital transformation in universities is not merely a technological issue but a complex socio-technical process requiring alignment between infrastructure, skills, organizational culture, and governance.

Thus, the results emerging from the literature review and the case study analysis provide an overview of the opportunities arising from the introduction of the DTs within university activities, but they also reveal several critical issues that must be addressed. Some of the most prevalent in the university environment concern the high costs of implementation and maintenance, the protection of privacy and the ethical use of collected data, the differences in socio-economic conditions among students, and the risk of compromising human value in implementing an excessively "digital" educational system.

By examining the digital transformation journey of Parthenope University and emphasizing the strategies adopted and the critical challenges encountered during implementation, the study proposes a conceptual framework to mitigate these risks and foster full technology integration in the university system. The establishment of partnerships, the introduction of guaranteed committees to ensure the ethical use of data and respect for privacy, the creation of dedicated study areas and the provision of both technological and financial resources to students, are some feasible solutions that could concretely foster the digital transformation of universities.

In this sense, this research offers theoretical contributions and practical implications.

From a theoretical perspective, this study contributes to the literature on digital transformation and strategic management

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in KIOs in several ways. First, it reinforces the view of digital transformation as a multi-dimensional and non-linear process, shaped by the interaction between technological capabilities, human capital, and institutional context. By empirically illustrating how DTs are often confined to basic applications, the study highlights the persistent gap between the transformative potential emphasized in the literature and the realities of implementation within universities. Second, the research advances existing studies by explicitly systematizing the drivers, mechanisms, and barriers of digital transformation in higher education into a conceptual framework. This framework integrates technological, organizational, cultural, and ethical dimensions, offering a more holistic understanding of why digital initiatives often fail to achieve their intended impact.

From a practical and managerial perspective, the findings suggest that, in the higher education sector, investments in digital infrastructure alone are insufficient to trigger meaningful transformation; continuous professional development emerges as a key condition for translating technological availability into effective teaching innovation. Second, the study highlights the importance of developing a coherent and shared digital strategy. Fragmented initiatives and isolated experiments limit the transformative potential of DTs. University managers should promote coordination across departments, encourage interdisciplinary experimentation, and align digital initiatives with broader institutional objectives related to teaching quality, student engagement, and organizational efficiency.

Although the paper has the typical limitations of the single case study approach (Yin 2009), it opens the way for further investigation and reflection.

First, the adoption of a single case study limits the generalizability of the findings. While the case allows for in-depth analysis and theory refinement, the results cannot be directly extended to all universities or national contexts. Future research could adopt multiple case studies or comparative designs to test and refine the proposed framework across different institutional settings.

Second, the study relies on data collected through questionnaires, which may be affected by subjective perceptions, social desirability bias, and respondents' individual interpretations of digital transformation. Although triangulation with documentary sources was employed to enhance reliability, further research could integrate additional methods, such as interviews, classroom observations, or longitudinal data, to capture behavioral changes over time.

Third, limitations also arise from the triangulation process itself. While multiple data sources were combined, the analysis primarily focused on professors' and students' perspectives, leaving other stakeholders—such as technical staff, university managers, and external partners—less explored. Future studies could adopt a broader stakeholder perspective to gain a more comprehensive understanding of digital transformation dynamics.

Finally, the cross-sectional nature of the study does not allow for an assessment of how digital transformation evolves over time. Longitudinal research could provide valuable insights into learning processes, path dependencies, and the long-term impact of digital strategies on teaching, research, and organizational performance.

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Ethics Statement

This research was conducted in accordance with the ethical standards of academic research.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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